

# TRIAD ERD Data Model

This document records the entity-relationship model behind TRIAD and the reasoning for its main design decisions. The model covers a single hospital and is aligned with FHIR R4. Its distinguishing feature is that every clinical fact carries its authorship and revision history as first-class structure.

## 1 Reading the diagram

| Symbol           | Meaning                                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Box              | Entity: something that exists on its own.                                                                                                    |
| Double box       | Weak entity: identified only together with its owner.                                                                                        |
| Diamond          | Relationship. A double diamond is the identifying relationship of a weak entity.                                                             |
| Oval             | Attribute. Ovals hanging off an oval form a composite attribute, such as Address.                                                            |
| Solid underline  | Key attribute.                                                                                                                               |
| Dashed underline | Partial key: identifies a weak entity together with its owner's key.                                                                         |
| Triangle (ISA)   | Subtype hierarchy, pointing toward the subtypes. Disjoint means each instance is exactly one subtype.                                        |
| Crow's feet      | Cardinality: two bars = exactly one; circle and bar = zero or one; circle and crow's foot = zero or many; bar and crow's foot = one or many. |
| Grey shading     | Marks TRIAD's contribution: authorship, revision, and source provenance.                                                                     |

## 2 Design principles

- Keys are real-world identifiers, never system-generated numbers. Where an entity has no natural key, it is modelled as a weak entity identified through its owner.
- No foreign keys appear at the conceptual level; all associations are expressed as relationships.
- Entity names are singular, and attribute names never repeat the entity name.
- An association that only exists when two entities meet is modelled as a relationship, following the Visit pattern between Patient and the clinical record.
- The diagram is acyclic: there is exactly one path between any two entities, so no fact can be reached by two routes that might disagree.

### 3 Keys, entity by entity

- **Patient** - composite key PHN + Name + Address, with attributes DateOfBirth, Sex, and PhoneNumber. Address is composite (Street, City, Province, PostalCode) because the province determines coverage in Canada. The combination reflects what the hospital verifies at registration; patients without a PHN are a known limitation, since a key attribute cannot be empty.
- **Staff** - key EmployeeNumber, the identifier the hospital assigns to every employee. It is stable and exists for every staff type, including a Medical Office Assistant who holds no licence. LicenseNumber is kept as a plain attribute on Doctor and Nurse; it cannot serve as the Staff key because an MOA has none.
- **Visit** - weak, owned by Patient through Has, with partial key StartDate. A patient cannot begin two encounters at the same instant. Class (in-person, phone, or virtual) lets phone calls and memos be modelled as visits.
- **ClinicalFact** - weak, owned by Visit through During, with partial key Code + RecordedAt. RecordedAt keeps repeated measurements of the same code within a visit distinct. A fact takes its clinical context and time from its visit, consistent with established clinical data models.
- **Assertion** - weak, owned by Staff through Asserts, with partial key AssertedAt. One person makes one assertion at one instant. About links an assertion to the fact it concerns without making the assertion depend on that fact.
- **Message** - composite key Channel + SentAt.
- **Department** and **Location** - key Name; within one hospital these are unique.
- **Medication** - key DIN, the Canadian Drug Identification Number.

### 4 Relationships and cardinalities

| Relationship                  | Reading                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------|
| Has (Patient-Visit)           | A patient has zero or many visits; each visit belongs to exactly one patient. Identifying.                     |
| During (Visit-ClinicalFact)   | A visit records zero or many facts; each fact is recorded during exactly one visit. Identifying.               |
| TakesPlaceIn (Visit-Location) | Many-to-many with StartDate and EndDate: a visit may move rooms, and a room hosts many visits.                 |
| WorksIn (Staff-Department)    | Each staff member works in one department; a department has zero or many staff. Carries StartDate and EndDate. |
| Asserts (Staff-Assertion)     | Each assertion has exactly one author; a staff member makes zero or many assertions. Identifying.              |

| Relationship                            | Reading                                                                                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| About<br>(Assertion-ClinicalFact)       | Each assertion concerns exactly one fact; each fact has one or more assertions, so no fact exists without someone claiming it.                                |
| Supersedes<br>(Assertion-Assertion)     | Unary: each assertion replaces at most one and is replaced by at most one.<br>The current fact is the assertion that nothing replaces, found in a single hop. |
| ExtractedFrom<br>(Assertion-Message)    | Each assertion comes from exactly one message; a message yields zero or many assertions.                                                                      |
| Prescribes<br>(Prescription-Medication) | Each prescription names one medication; a medication appears in many prescriptions.                                                                           |

## 5 Why the model is acyclic

Three associations are deliberately omitted so that the diagram stays a tree:

- No direct Patient-ClinicalFact link: a fact reaches its patient only through its visit.
- No Staff-Visit link: participation in a visit is recovered from who asserted facts during it. This also rules out a Department-Location link, which would close a loop back to Staff.
- No links between ClinicalFact subtypes, which would close a loop through the shared parent.

A shared Person supertype over Staff and Patient is also omitted, because the two have different keys and a common parent would give Patient a second path into the model.

## 6 Authorship and revision (the assertion pipeline)

TRIAD's contribution is recording, for every clinical fact, who asserted it and what it superseded.

- A Message arrives: an email, memo, or phone note.
- Clinical facts are extracted, each recorded during a visit of the appropriate class.
- For each fact an Assertion records the human author, their role, and the time. When the work is done by an agent, it acts on behalf of that human author, who remains responsible; the agent is never itself the asserter.
- When a new assertion changes an earlier one, the two are linked by Supersedes.

The current state of the record is the set of assertions that nothing supersedes; history is read by following Supersedes backward, and every step names a person and a source message.

## 7 FHIR R4 alignment

| TRIAD                      | FHIR R4 resource                                                                                                                                                                                             |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Patient                    | Patient                                                                                                                                                                                                      |
| Visit                      | Encounter                                                                                                                                                                                                    |
| Location                   | Location                                                                                                                                                                                                     |
| Department                 | Organization                                                                                                                                                                                                 |
| Staff (Doctor, Nurse, MOA) | Practitioner / PractitionerRole                                                                                                                                                                              |
| Diagnosis                  | Condition                                                                                                                                                                                                    |
| Procedure                  | Procedure                                                                                                                                                                                                    |
| Allergy                    | AllergyIntolerance                                                                                                                                                                                           |
| Observation                | Observation                                                                                                                                                                                                  |
| LabOrder                   | ServiceRequest                                                                                                                                                                                               |
| LabReport                  | DiagnosticReport                                                                                                                                                                                             |
| Prescription               | MedicationRequest                                                                                                                                                                                            |
| Medication                 | Medication                                                                                                                                                                                                   |
| Message                    | Communication                                                                                                                                                                                                |
| Assertion, Supersedes      | FHIR supports authorship and supersession only optionally, through fields such as recorder and asserter and links such as Patient.link and priorPrescription. TRIAD makes both structural across every fact. |